SQL Project Report

Exploratory Data Analysis using MySQL — a full year of pizza sales data analyzed through SQL queries, joins, aggregations, and window functions.

SQL | WINDOW FUNCTIONS | JOINS | AGGREGATIONS | EDA

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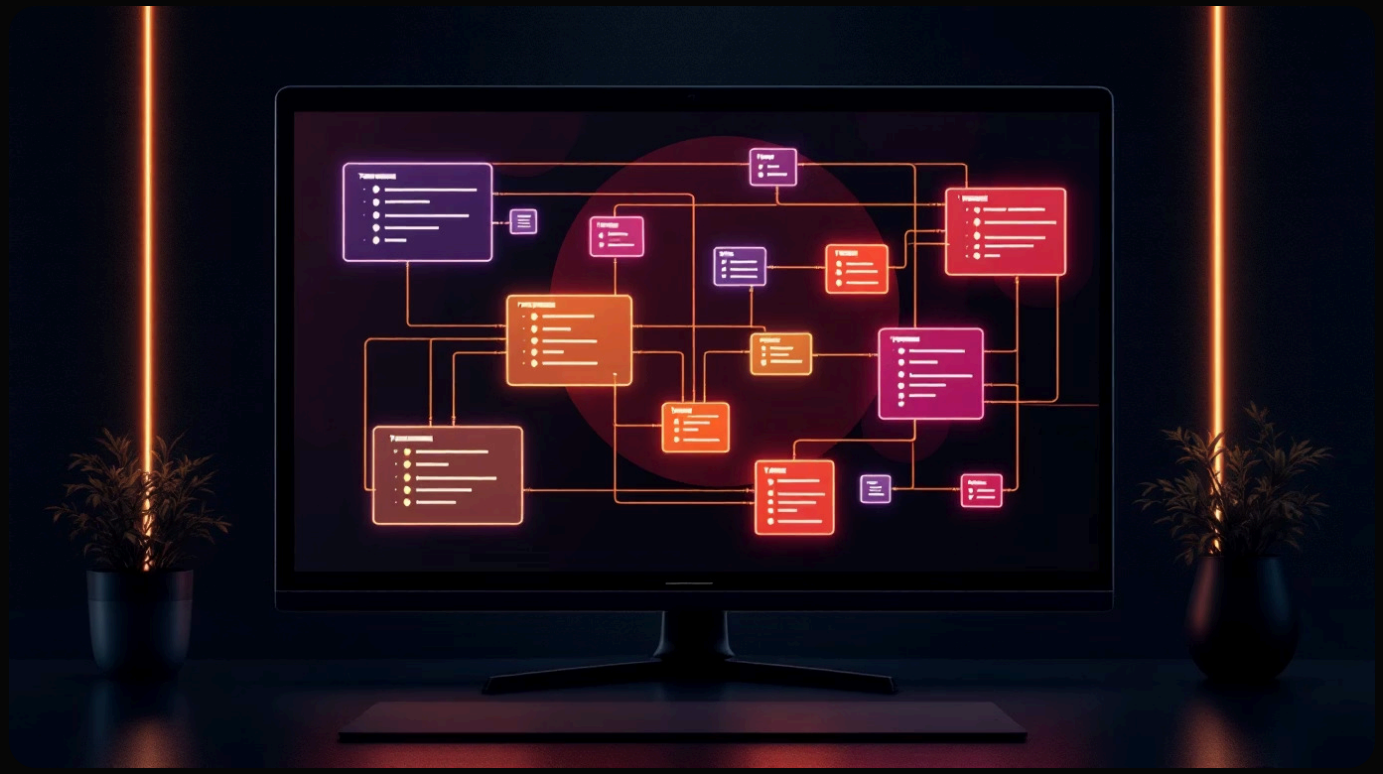
Project Overview



An end-to-end EDA on a pizza restaurant's sales database using MySQL. The dataset captures a full year of transactional data across four relational tables, extracting insights on revenue, order volumes, and customer ordering patterns.

Key Objectives

- Quantify overall business performance (orders, revenue)
- Identify best-selling products and high-value items
- Analyze customer ordering behavior across time and size
- Support data-driven decisions for menu and operations optimization

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Database Schema

Four tables linked via foreign keys form the relational backbone of this analysis.

Table	Columns	Description
orders	order_id, order_date, order_time	One row per customer order
order_details	order_details_id, order_id, pizza_id, quantity	Line items linking orders to pizzas
pizzas	pizza_id, pizza_type_id, size, price	SKU-level pizza info with pricing
pizza_types	pizza_type_id, name, category, ingredients	Pizza names, categories & ingredients

order_details references orders (order_id) and pizzas (pizza_id); pizzas references pizza_types (pizza_type_id).

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Key Metrics at a Glance

21,350

Total Orders

Orders processed over the analyzed period

$817K

Total Revenue

Generated from pizza sales

32

Pizza Types

Unique pizza varieties on the menu

5

Sizes

Sizes available: S, M, L, XL, XXL

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Revenue & Order Analysis

Total Revenue

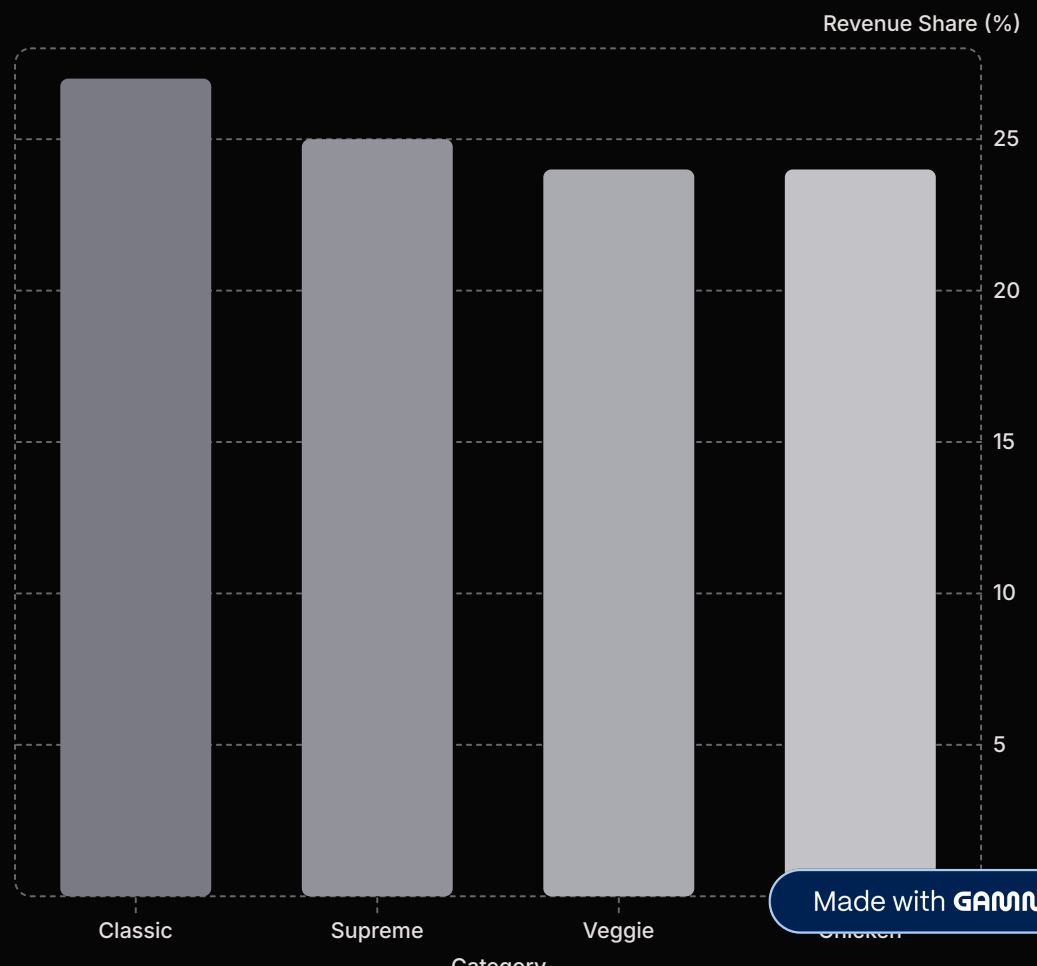
$817,860.05 generated across all orders — averaging **~$38.31 per order**.

Highest-Priced Pizza

The Greek Pizza at **$35.95** — the premium offering targeting gourmet customers.

Average Daily Volume

138 pizzas ordered per day on average — a key baseline for prep and staffing.



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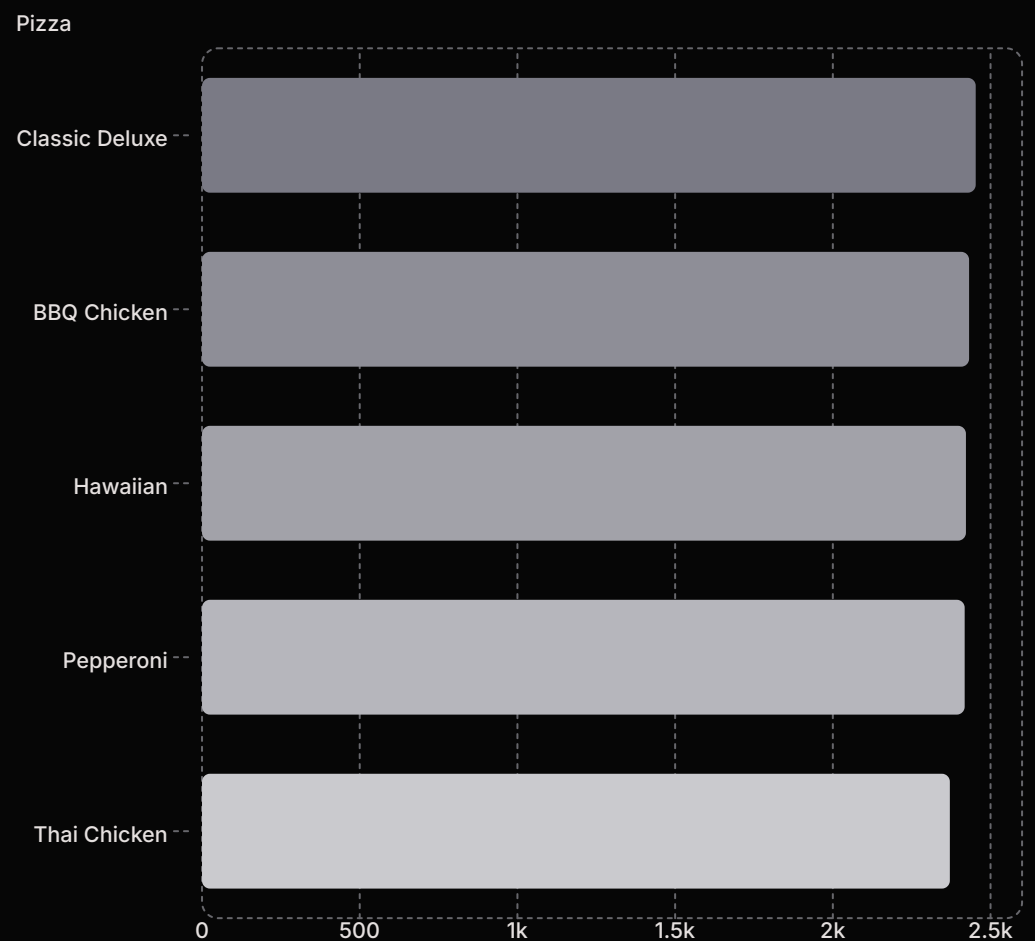
Top 5 Most Ordered Pizzas



Tight Competition

Classic Deluxe leads by quantity, but the top 5 are **tightly clustered** — suggesting balanced customer preferences across popular options.

Category Distribution

Classic pizzas dominate with **~30%** of all quantities sold. The remaining three categories are fairly even, leaving room to promote Chicken and Veggie options.

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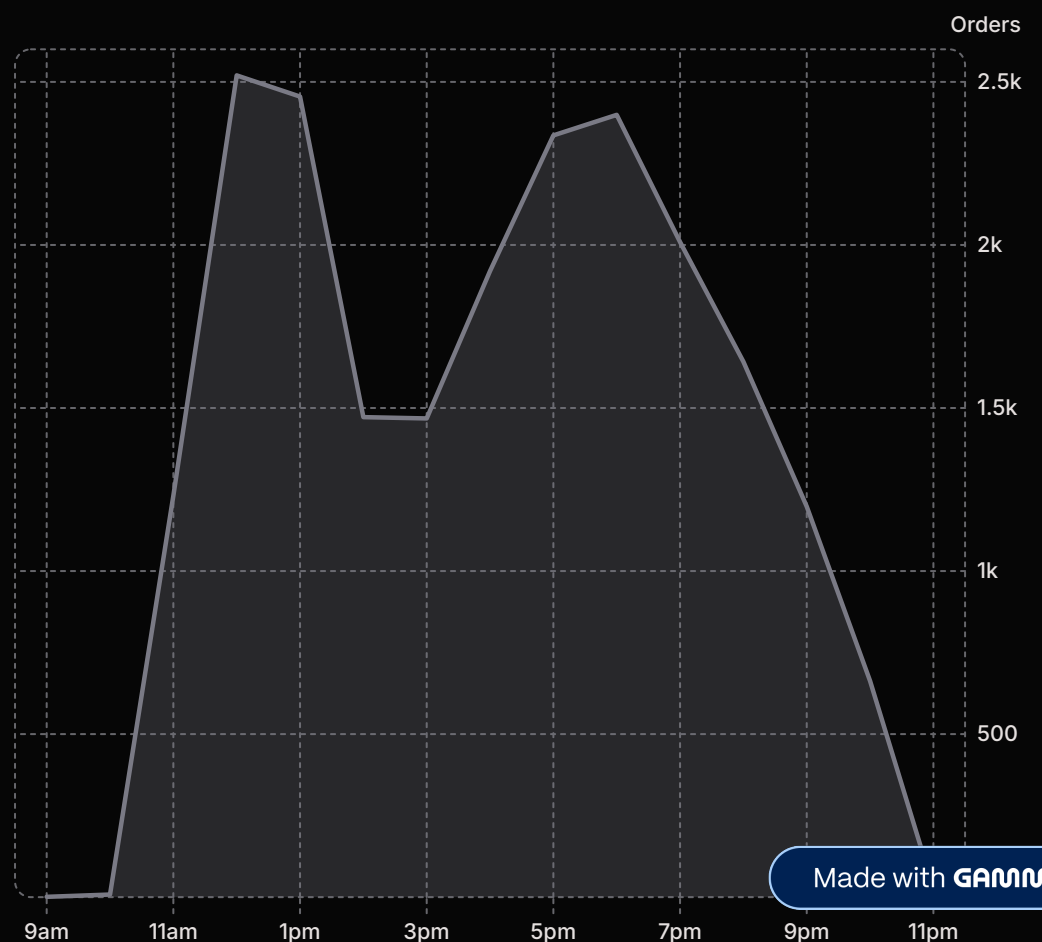
Order Patterns: Size & Time

Most Common Size

Large (L) leads with **18,526 orders** — nearly 37% of all orders — followed by Medium and Small. XL and XXL are niche sizes.

Peak Ordering Hours

Two clear rushes: **12:00–13:00** (lunch) and **17:00–18:00** (dinner). Staff scheduling and prep should align with these windows.



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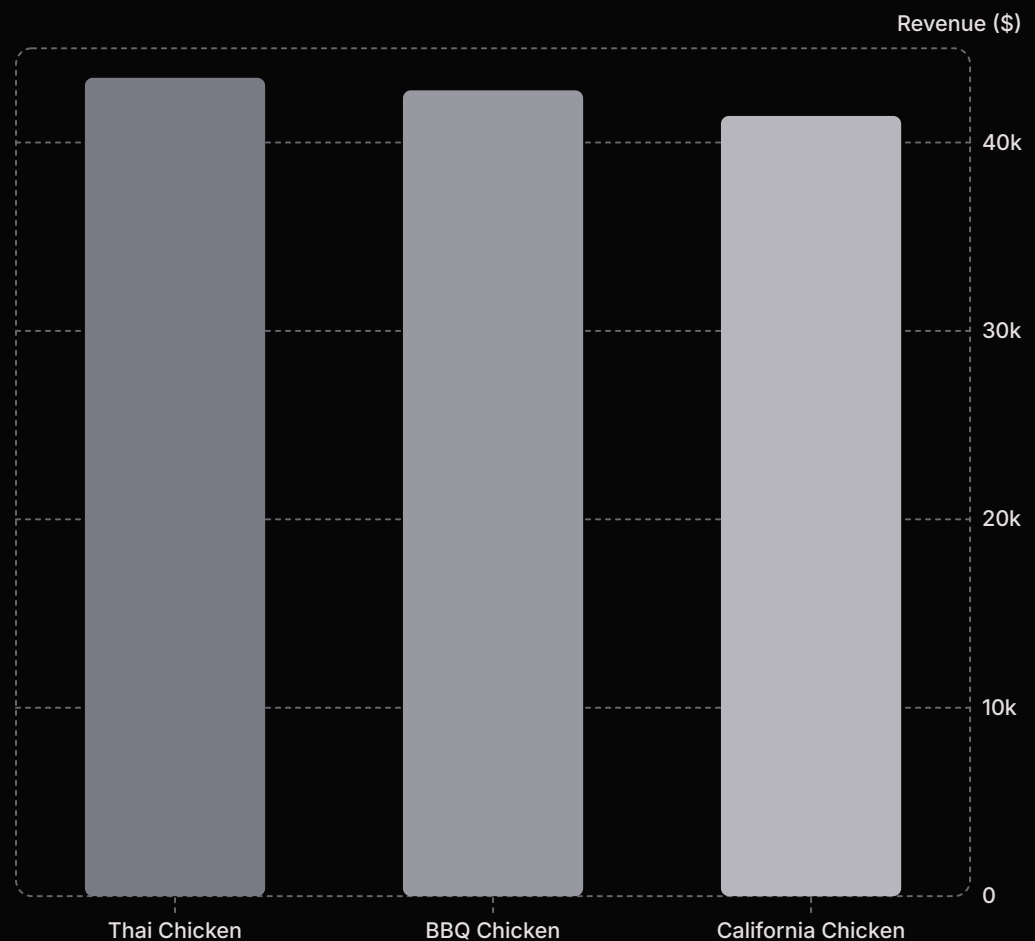
Revenue Leaders by Category



Chicken Dominates Revenue

All three top revenue-generating pizzas are **Chicken variants** — despite Chicken being the **smallest category** by menu size (only 6 items). A strong signal to expand this category.

Revenue Balance

Overall revenue is remarkably balanced: Classic 27%, Supreme 25%, Veggie 24%, Chicken 24% — a spread of only **3%**, reflecting a diverse and healthy menu mix.

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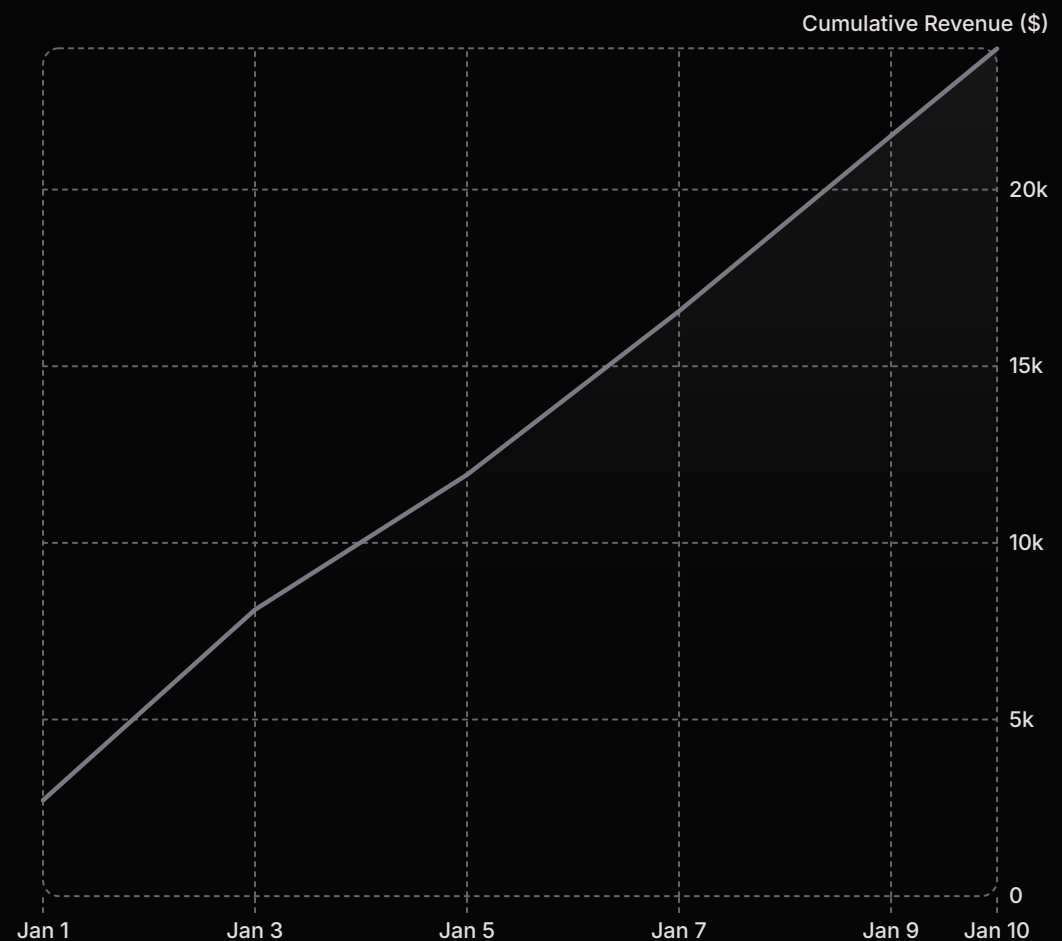
Cumulative Revenue Trend



Window Function in Action

`SUM(...) OVER(ORDER BY order_date)` computes a running total without collapsing rows — a powerful window function technique.

Steady Growth

Revenue grows from **$2,713 on Jan 1** to **$817,860 by year end** with no major dips, indicating consistent year-round demand.

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Key Business Insights

Chicken is the Revenue King

Smallest category (6 items) yet occupies all top 3 revenue spots — expand this menu.

Balanced Category Mix

Revenue spread is only 3% across all four categories — no over-dependence on any single type.

Size Upsell Opportunity

Large pizzas dominate orders. Upsell campaigns from Medium to Large could meaningfully boost revenue.

Peak Hour Optimization

Two clear peaks at lunch (12–13h) and dinner (17–18h). Introduce off-peak discounts (15–16h) to smooth demand.

Premium Potential

The Greek Pizza ($35.95) occupies the luxury tier. Marketing it as a premium/sharing option could raise average order value.

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